Exploratory Data Analysis on Diwali Sales Dataset

Project Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Diwali Sales dataset to understand customer buying behavior, popular product categories, and sales trends during the Diwali season.

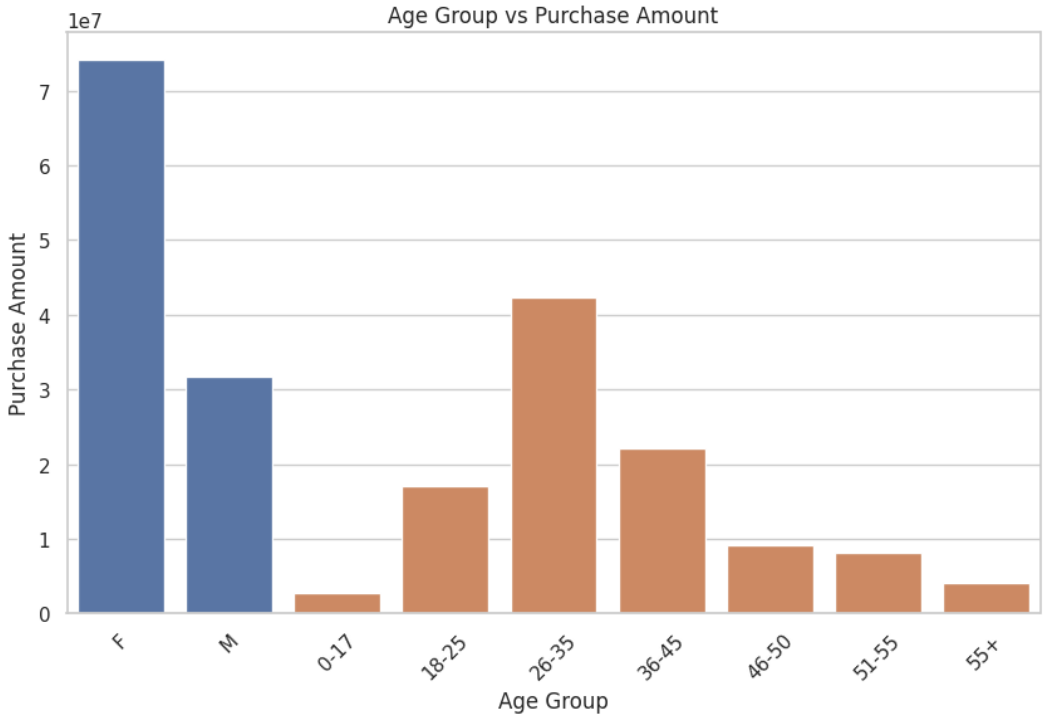
Tools and Libraries Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn

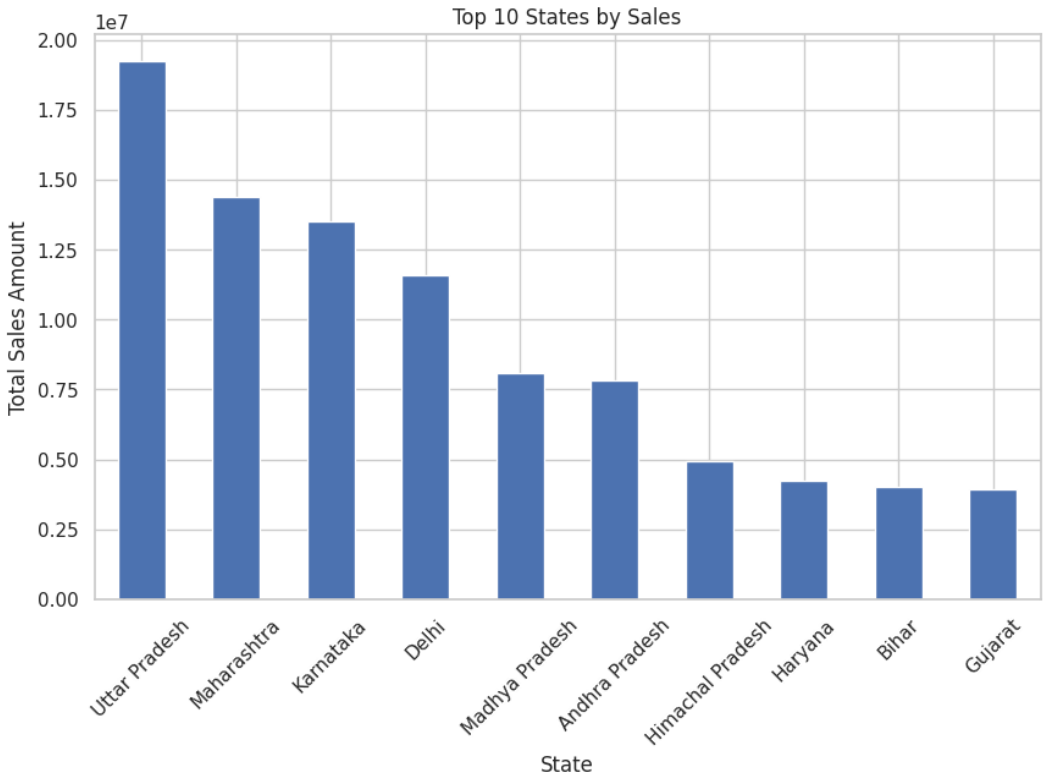
Screenshots of Key Visualizations

The following screenshots were taken from the Google Colab notebook used in this project.

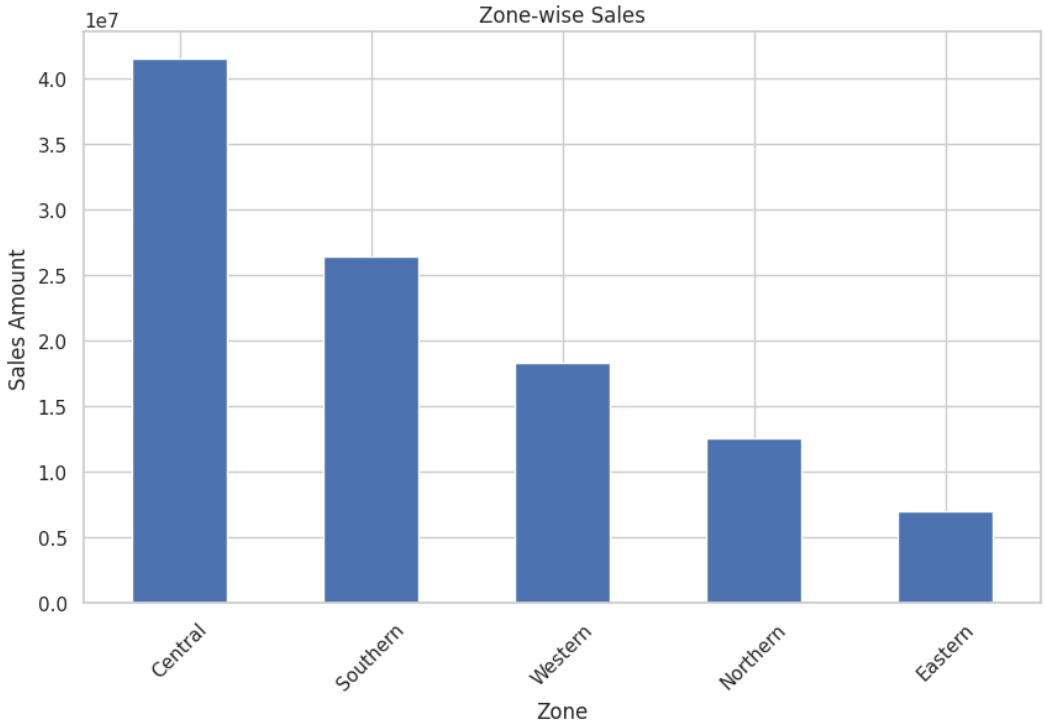
Visualization 1



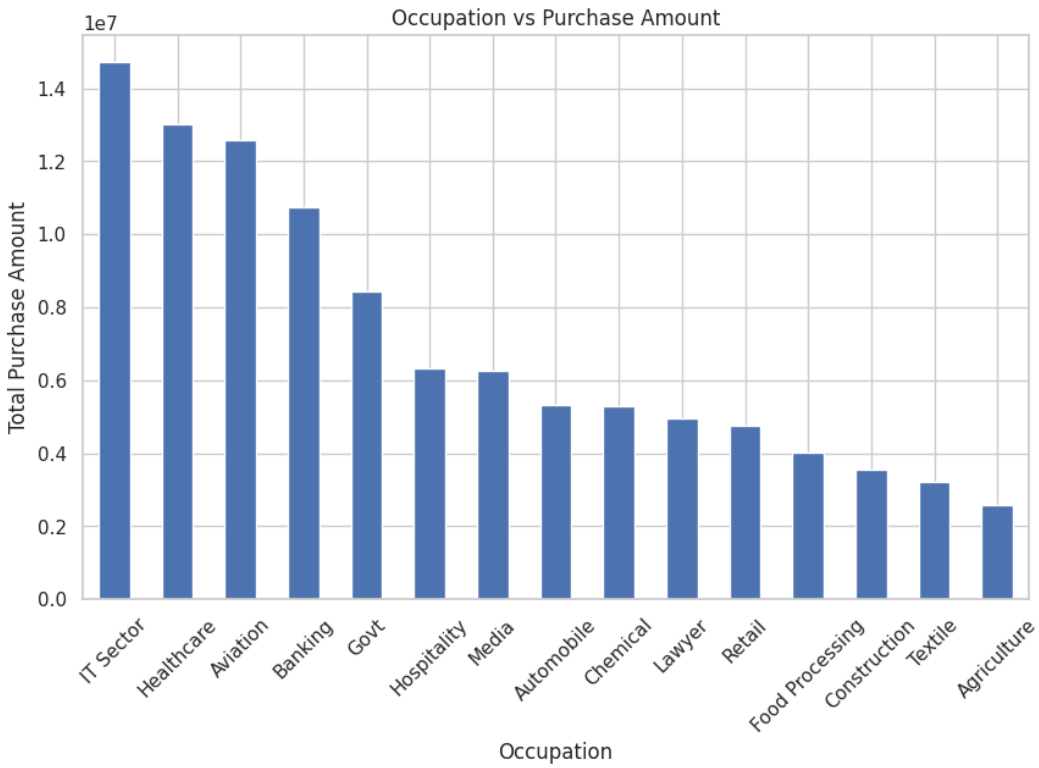
Visualization 2



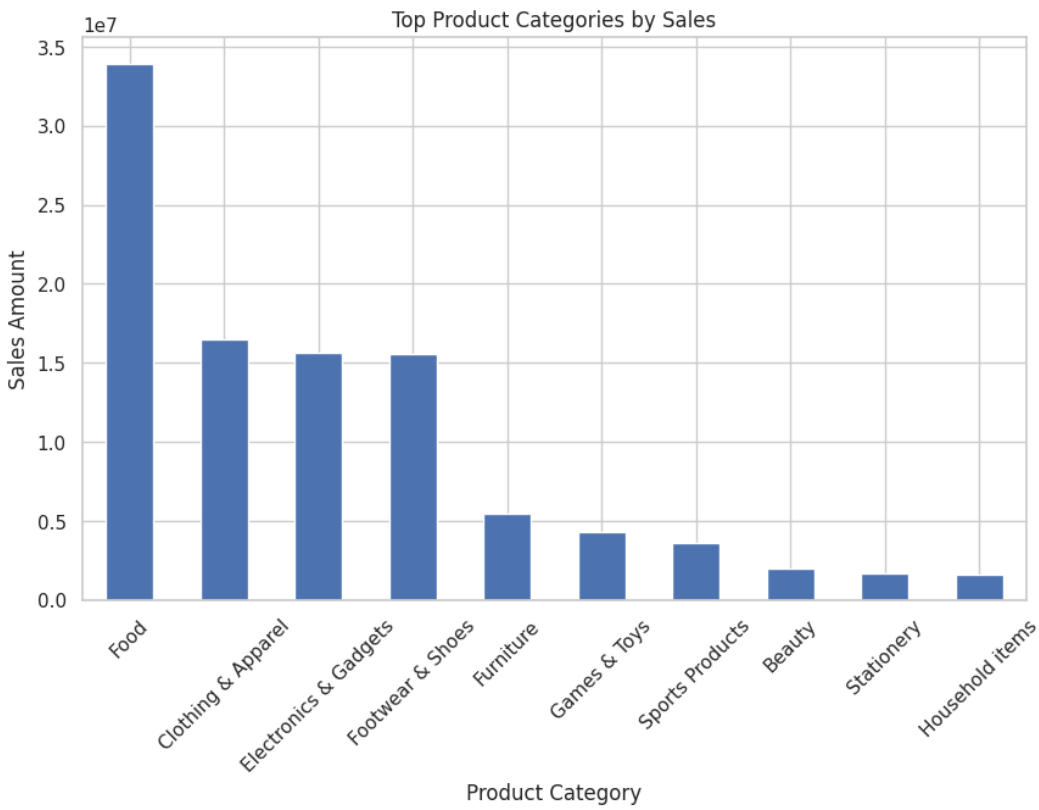
Visualization 3



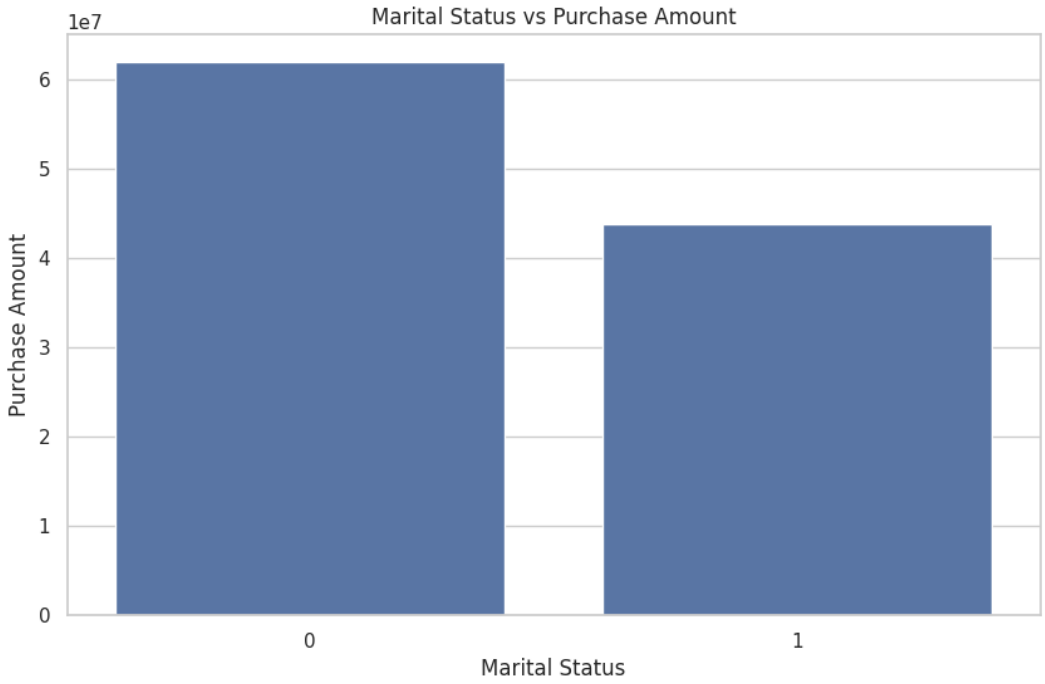
Visualization 4



Visualization 5



Visualization 6



Summary of Insights

- Female customers contributed more to total purchases than male customers.
- Customers aged 26–35 made the highest number of purchases.
- States such as Uttar Pradesh, Maharashtra, and Karnataka had the highest sales.
- Food, Clothing, and Electronics were among the most popular product categories.
- Working professionals and IT employees had higher purchase amounts.
- Most purchases were in the medium price range.

Challenges Faced

- Missing values in some columns
- Duplicate rows in the dataset
- Incorrect data types in the Amount column
- Outliers in purchase amount
- City column not available in the dataset

Solutions Applied

- Removed missing values using dropna()
- Removed duplicate records
- Converted Amount column into numeric format
- Removed outliers using IQR method
- Used Zone-wise analysis instead of City-wise analysis

Conclusion

This project successfully analyzed customer behavior and sales trends during the Diwali season. The findings can help businesses improve marketing strategies, manage inventory, and target customers more effectively.